

Zhongyang Li

601-666-0859 | li.17258@osu.edu | [zhongyangli0124.github.io](https://github.com/zhongyangli0124)

Research Interests: AI Agents, Mixture-of-Experts, Test-Time Adaptation

EDUCATION

The Ohio State University

PhD in Computer Science

Columbus, US

Aug. 2026 – Present

Johns Hopkins University

Master of Science in Computer Science | GPA:3.6

Baltimore, US

Sep. 2023 – May 2025

Dalian University of Technology

Bachelor of Science in Software Engineering | GPA:3.8

Dalian, China

Sep. 2019 – Jun. 2023

PUBLICATIONS

[1] QUEST: Training Frontier Deep Research Agents with Fully Synthetic Tasks

NeurIPS 2026

Open deep research agents (2B–35B) trained on fully synthetic long-horizon tasks via mid-training, SFT, and RL.

Jian Xie, Tianhe Lin, Zilu Wang, ..., [Zhongyang Li](#), ..., Yu Su, Huan Sun

[2] Routing Entropy: A Hidden Self-Verifier for Free in Mixture-of-Experts LLMs

COLM 2026

Shows MoE routing entropy is a free, training-free self-verifier of output correctness.

[Zhongyang Li](#), Ziyue Li, Tianyi Zhou

[3] Routing Manifold Alignment Improves Generalization of MoE LLMs

ICLR 2026

Aligns routing-weight manifolds with task embeddings via light router fine-tuning; 5–15% accuracy gains.

[Zhongyang Li](#), Ziyue Li, Tianyi Zhou

[4] C3PO: Critical-Layer, Core-Expert, Collaborative Pathway Optimization in LLMs

COLM 2025

Test-time optimization of critical-layer, core-expert pathways, improving MoE LLM accuracy by 7–15%.

[Zhongyang Li](#), Ziyue Li, Tianyi Zhou

[5] R2-T2: Re-Routing in Test-Time for Multimodal Mixture-of-Experts

ICML 2025

Test-time re-routing for multimodal MoEs, achieving 5–10% accuracy gains across benchmarks.

[Zhongyang Li](#), Ziyue Li, Tianyi Zhou

RESEARCH EXPERIENCE

Research Intern - University of Maryland, College Park

Jun. 2024 – Apr. 2026

Advised by Prof. Tianyi Zhou

College Park, US

- Research on MoE routing in LLMs/LMMs via test-time optimization and router fine-tuning, leading to four first-author papers [2–5]; enabled smaller MoE models to rival or surpass larger dense models.

Research Intern - Johns Hopkins University

Sep. 2024 – Dec. 2024

Advised by Prof. Ashutosh Dutta

Baltimore, US

- Hybrid LLM watermark detection combining token-group monitoring with Lomb–Scargle frequency analysis; high detection rates on GPT/LLaMA with <1% BLEU degradation.

SERVICE

Reviewer: ICLR 2026, ICML 2026, NeurIPS 2026, COLM 2026, ICLR 2027

Conference Volunteer: COLM 2025

SKILLS

Agents: Tool use / function calling, MCP, multi-agent systems, agent memory, GUI & web agents

Training & Eval: SFT, agentic RL (GRPO/PPO, verl), synthetic trajectory generation, LLM-as-a-judge

Infrastructure: Python/PyTorch, vLLM, SGLang, DeepSpeed/FSDP, Docker/Apptainer, Slurm, Linux, Git